A Test of Mean Resolution Time

Unit 4 · Topic 4.5 · THE PROBLEM

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

Leah: "The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide."

Damon: "Because $p > 0.05$, accept H_0. The test proves the population mean was exactly 12 hours and the observed difference was only chance."

Ticket sample

Summary	Value
Population	1200
n	24
Mean (h)	11.1
SD (h)	2.8
Shape	Symmetric; no outliers

FIRST TAKE — one sentence before you work the parts

Does this small study seem able to prove that the average was exactly 12 hours? Explain in one sentence.

DOK 1 **(a)** State H_0 and H_a for the population mean resolution time.

DOK 2 **(b)** Using $\bar{x} = 11.1$, $s = 2.8$, and $n = 24$, calculate t. Use the supplied $p = 0.0645$; you do not need to find it again.

DOK 3 **(c)** In 3–4 sentences, decide whose account is justified. Compare the p-value with 0.05, state the conclusion for all 1,200 tickets, and explain why failure to reject does not prove equality or that chance was the only explanation.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.